Module 04: Cognition — Search Engines and Algorithms

Learning Objectives

1. Understand that **Cognition** for a computer is running a **Algorithm**.
2. Learn how a **Search Engine** finds what you are looking for.
3. Know that websites try to predict what you want to see (Recommendations).

Introduction: The Librarian of the Internet

Imagine a library with 1 Billion books. They are all in a big pile on the floor. If you want a book about "Cats," how do you find it? You need a Librarian. Google/Bing are the Librarians. They have an index of everything. When you ask, they find it in 0.0001 seconds.

Key Concepts

1. The Algorithm (A Recipe)

An Algorithm is just a fancy word for a Recipe.

1. Crack egg.
2. Mix.
3. Cook. Computer Algorithm:
4. Look at search word.
5. Find pages with that word.
6. Show the most popular one first.

2. Recommendation (The Prediction)

Why does YouTube show you videos of Minecraft? Because you watched Minecraft yesterday! The computer creates a Model of YOU. "He likes blocks. Show more blocks." It is trying to predict your future behavior.

3. Filter Bubbles

If the computer ONLY shows you what you like, you never see new things. It's like only eating pizza forever. Sometimes you need to search for something totally different!

Activities

Activity 1: The Sorting Algorithm

Sort a deck of cards. Method A: Throw them on the floor and pick them up one by one. (Inefficient). Method B: Separate red and black first. Then suits. Then numbers. (Efficient). Computers use efficient sorting to be fast.

Activity 2: The "Autofill" Game

Start a sentence: "I want to eat..." Let your phone finish it using the middle button. "I want to eat... pizza... tomorrow... with... Mom." The phone is guessing the next word using probability!

Summary

Computers are fast thinkers, but they only know what we teach them. You are the driver; the computer is the map.

References

- *Computational Fairy Tales* by Jeremy Kubica
- *Thinking, Fast and Slow* (Concepts adapted for kids)